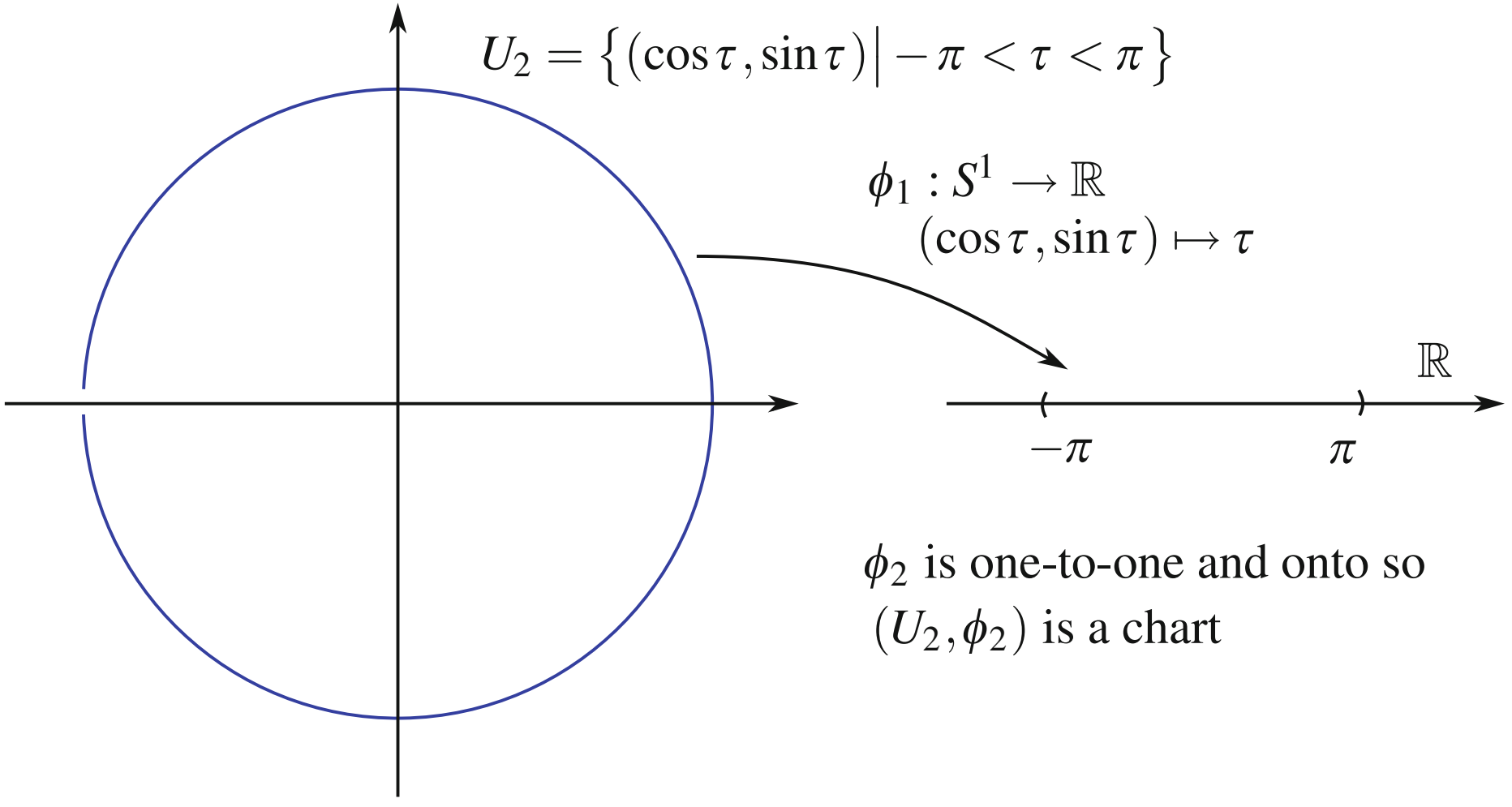

$U_2 = \{(\cos\tau, \sin\tau) \mid -\pi < \tau < \pi\}$
$\phi_1 : S^1 \to \mathbb{R}$
$(\cos\tau, \sin\tau) \mapsto \tau$
$\mathbb{R}$
$-\pi$
π
ϕ_2 is one-to-one and onto so
(U_2, ϕ_2) is a chart